EXP NO:12 EXPERIMENT TO DETECT OUTLIERS IN A GIVEN DATA SET

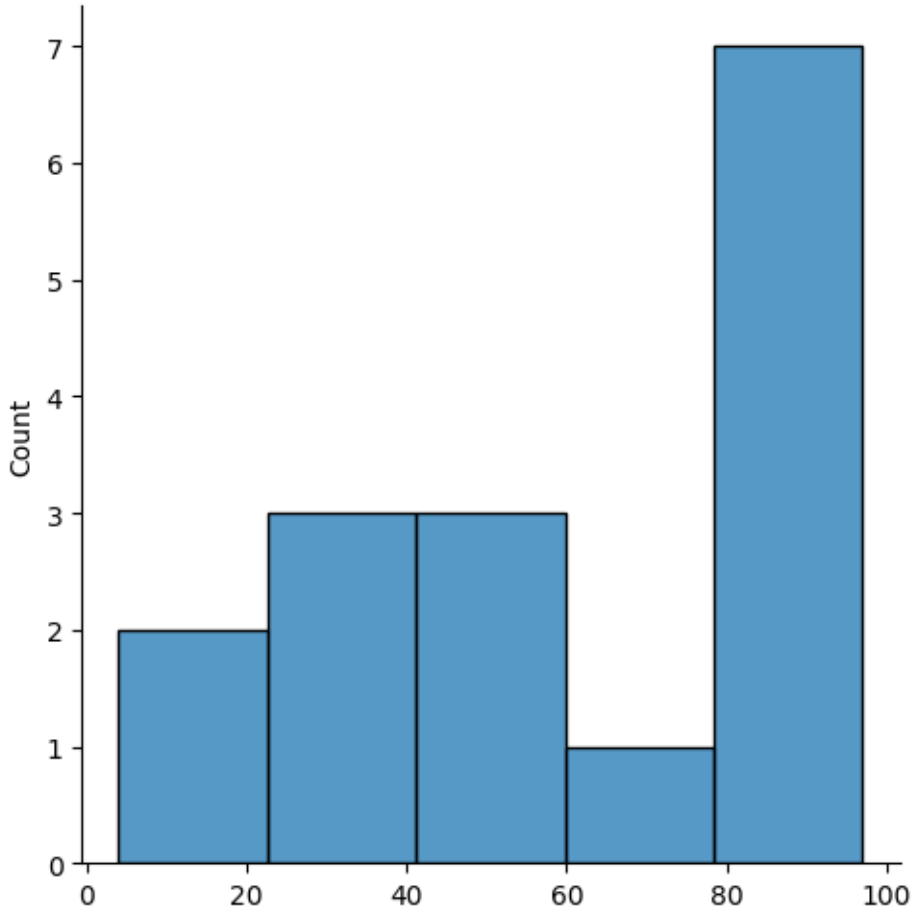
CODE :

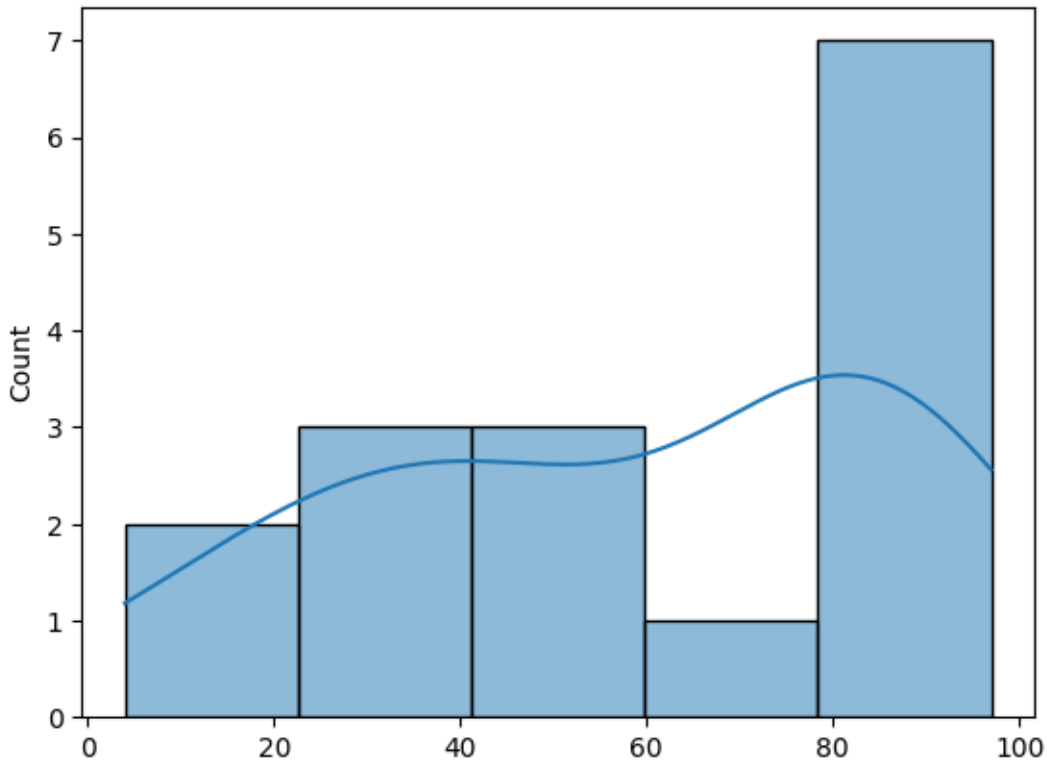
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
array = np.random.randint(1, 100, 16)
print(array)
print(array.mean())
print(np.percentile(array, 25))
print(np.percentile(array, 50))
print(np.percentile(array, 75))
print(np.percentile(array, 100))
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur
lr, ur = outDetection(array)
print(lr, ur)
sns.displot(array)
plt.show()
```

```
sns.histplot(array, kde=True)  # updated instead of distplot
plt.show()
new_array = array[(array > lr) & (array < ur)]
print(new_array)
sns.displot(new_array)
plt.show()
lr1, ur1 = outDetection(new_array)
print(lr1, ur1)
final_array = new_array[(new_array > lr1) & (new_array < ur1)]
print(final_array)
sns.histplot(final_array, kde=True)
plt.show()
```

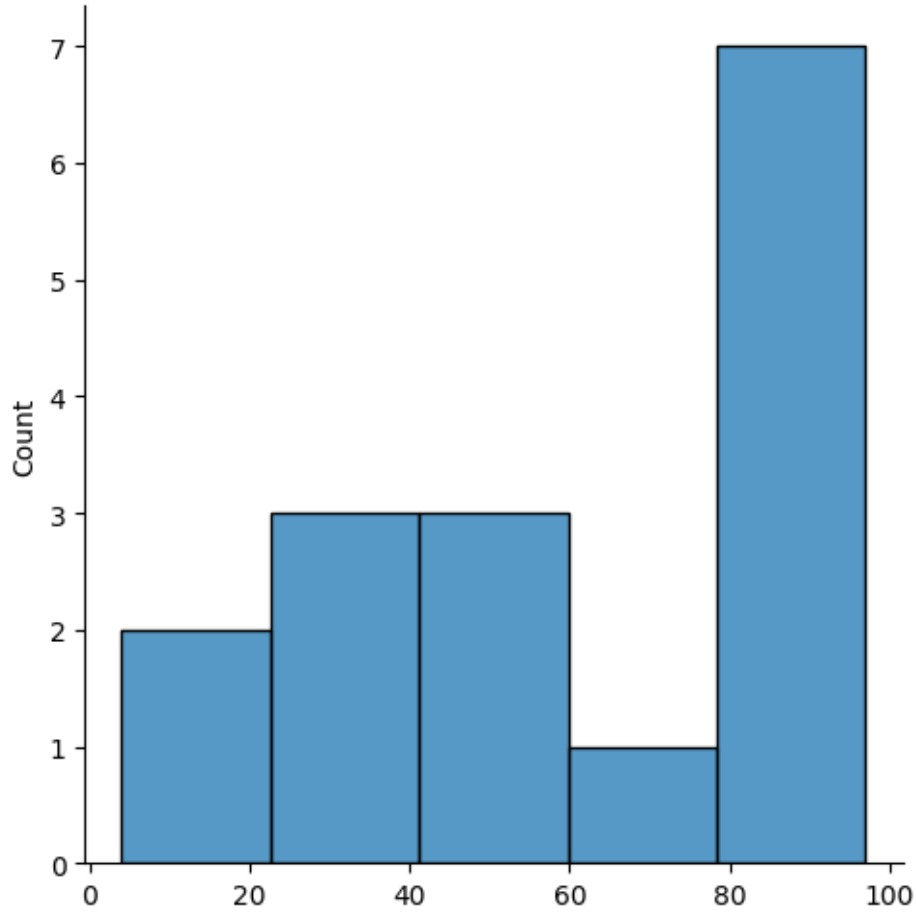
OUTPUT :

```
[23 78 52 21 39 46  4 97 47 80 79 84 82 84 29 89]
58.375
36.5
65.0
82.5
97.0
-32.5 151.5
```



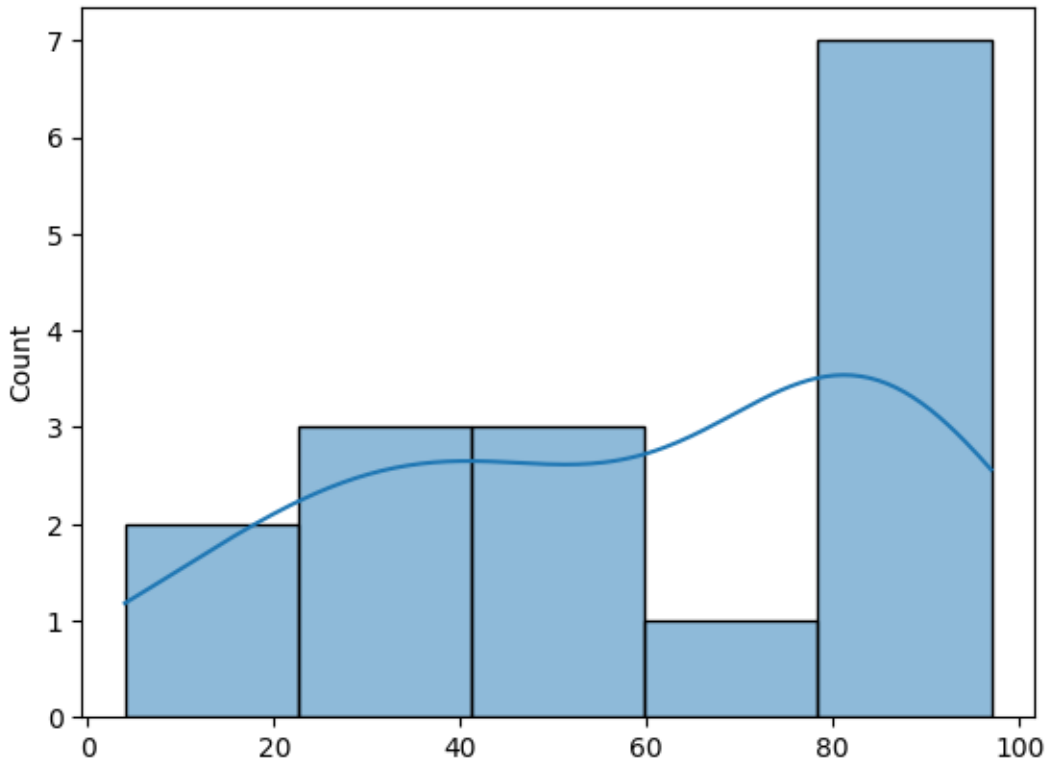


[23 78 52 21 39 46 4 97 47 80 79 84 82 84 29 89]



-32.5 151.5

[23 78 52 21 39 46 4 97 47 80 79 84 82 84 29 89]


Count
0
1
2
3
4
5
6
7
0
20
40
60
80
100